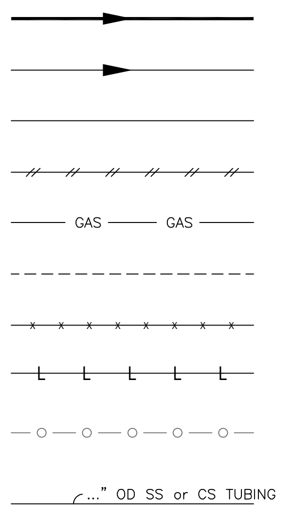
ABBREVIATIONS

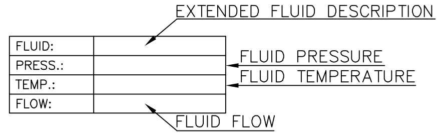
AS	–AIR SUPPLY
ATM	–ATMOSPHERE
AOUT	–AUTOMATIC OUTPUT
A/G	–ABOVE GROUND
BD	–BLOWDOWN
BDV	–EMERGENCY BLOWDOWN VALVE
BL	–BATTERY LIMIT
BTL	–BOTTOM TANGENT LINE
CA	–CORROSION ALLOWANCE
CKV	–CHECK VALVE
C	–COMPRESSOR
CG	–COUPLING GUARD
CP	–CORROSION PROBE
CPL	–COUPLING
CM	–COMPRESSOR MOTOR
CS	–CARBON STEEL
CSC	–CAR SEAL CLOSED
CSO	–CAR SEAL OPEN
CSS	–CONTROL AND SAFETY SYSTEM
CT	–CAPILLARY TUBE
CUST	–CUSTOMER
DCV	–DIRECTIONAL CONTROL VALVE
DB	–DEAD BAND
DP	–DESIGN PRESSURE
DS	–DISCHARGE SILENCER
DT	–DESIGN TEMPERATURE
DIR	–DIRECT ACTING
EL	–ELEVATION
ESD	–EMERGENCY SHUTDOWN
FB	–FULL BORE
FC	–FAIL CLOSED
FI	–FAIL INDETERMINATE
FO	–FAIL OPEN
FLP	–FAIL LAST POSITION
FLT	–FILTER
FD	–FLOAT DRAINER
FDI	–FLAME DETECTOR INDICATOR
FDT	–FLAME DETECTOR TRANSMITTER
FDR	–FILTER DRIER
FM	–FLOW METER
FX	–FROST EXTENSION
FV	–FULL VACUM
GDI	–GAS DETECTOR INDICATOR
GDT	–GAS DETECTOR/TRANSMITTER
GOV	–GAS OPERATED VALVE
HC	–HEAT CONSERVATION
HCV	–HAND CONTROL VALVE
HI	–HIGH
HOA	–HAND–OFF–AUTO SWITCH
HMI	–HUMAN MACHINE INTERFACE
HMS	–HAND MOMENTARY SWITCH
HPV	–HIGH POINT VENT
HR	–HANDLE REMOVE
HV	–HAND VALVE
HX	–HEAT EXCHANGER
HTR	–HEATER
HS	–HAND SWITCH
HW	–HAND HOLE / HAND WAY
IA	–INSTRUMENT AIR SUPPLY
ID	– INTERNAL DIAMETER
IG	–INSTRUMENT GAS SUPPLY
I/O	– INPUT / OUTPUT
IE	–CURRENT TRANSFORMER
IES	–INDUSTRIAL EQUIPMENT SOLUTION
LC	–LOCKED CLOSED
LO	–LOCKED OPEN
LNGP	–LNG GPS
LS	–LIQUID SEAL
LTCS	–LOW TEMPERATURE CARBON STEEL
LPD	–LOW POINT DRAIN
MAX	–MAXIMUM
MAWP	–MAXIMUM ALLOWABLE WORKING PRESSURE
MDMT	–MINIMUM DESIGN METAL TEMPERATURE
MCC	–MOTOR CONTROL CENTER
MIN	–MINIMUM
MMS	–MACHINE MONITORING SYSTEM
MOUT	–MANUAL OUTPUT
MOV	–MOTOR OPERATED VALVE
MW	–MANWAY
MS	–MOTOR STARTER
NC	–NORMALLY CLOSED
NNF	–NORMALLY NO FLOW
NLL	–NORMAL LIQUID LEVEL
NO	–NORMALLY OPEN
OIC	–OPERATOR INTERFACE COMPUTER
OAH	–OVERALL HEIGHT
P	–PIPE
PB	–PUSH BUTTON
PL	–PANEL/PILOT LIGHT
PLC	–PROGRAMMABLE LOGIC CONTROLLER
PSS	–PLANT SAFETY SHUTDOWN
PSV	–PRESSURE SAFETY VALVE (RELIEF VALVE)
P	–PUMP
PM	–PUMP MOTOR
PP	–PERSONAL PROTECTION
PRV	–PRESSURE RELIEF VALVE
RB	–REDUCED BORE
REV	–REVERSE ACTING
RFLX	–REFLEX GAUGE
RO	–RESTRICTION ORIFICE
RFWN	–RAISED FACE WELD NECK FLANGE
RFSO	–RAISED FACE SLIP ON FLANGE
RFSW	–RAISED FACE SOCKET WELD FLANGE
RST	–RESET (INTEGRAL)
RTD	–RESISTANCE TEMPERATURE DEVICE
SC	–SAMPLE CONNECTION
SD	–SHUT DOWN
SDV	–SHUT DOWN VALVE
S/F	–SEAM TO FACE OR FLANGE
SG	–SIGHT GLASS
SK	–SURFACE MOUNTED (SKIN)
SM–SM	–SEAM TO SEAM
SO	–STEAM OUT
SP	–SET POINT
SS	–SHELL SIDE, STAINLESS STEEL, SELECTION SWITCH
S/S	–SEAM TO SEAM
STR	–STRAINER
T	–TUBING
TC/K	–THERMOCOUPLE/(TYPE)
T/L	–TUBE LENGHT
TS	–TUBE SIDE
TSO	–TIGHT SHUT OFF
TS/TS	–TUBE SHEET TO TUBESHEET
TSV	–THERMAL RELIEF VALVE
T/T	–TANGENT TO TANGENT
TW	–THERMOWELL
UC	–UTILITY CONNECITON
UCP	–UNIT CONTROL PANEL
U/G	–UNDERGROUND
XA/UA	–COMMON ALARM/SHUTDOWN
V	–VESSEL
CWS	–COOLING WATER SUPPLY
CWR	–COOLING WATER RETURN
XT	–INTERFACE MODULE FOR ACCELEROMETER
XV	–ON–OFF VALVE (DIRECT ACTUATED OR PILOT OPERATED)
V/H	–VENT HEADER
VTA	–VENT TO ATMOSPHERE
VTG	–VENT TO GRADE
VTH	–VENT TO HEADER
V/R	–VENT/RELIEF
ZT	–PROXIMITOR/POTENTIOMETER
ZE	–AXIAL PROBE/POTENTIOMETER
ZI	–POSITION INDICATOR/AXIAL DISPLACEMENT INDICATION

LINE SYMBOLS

MAIN PROCESS LINE
SECONDARY PROCESS LINE
INSTRUMENT LINE
PNEUMATIC SIGNAL
GAS SIGNAL
ELECTRIC SIGNAL
CAPILLARY TUBE
HYDRAULIC SIGNAL
INTERNAL SYSTEM LINK
(SOFTWARE OR DATA LINK)
TUBING LINE



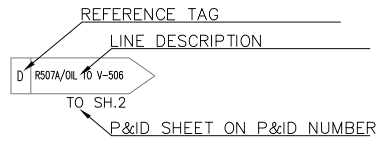
LINE OPERATING CONDITIONS



EXAMPLE

FLUID:	R507a SUBC. LIQUID
PRESS.:	183.8 psiA
TEMP.:	108.7F
FLOW:	54341 lb/h (90 GPM)

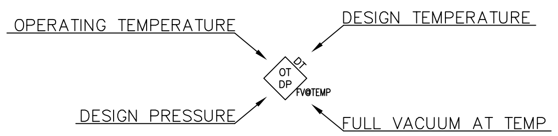
P&ID PROCESS CONTINUATION ARROW



MISCELLANEOUS

Slope
Blind Flanges
Concentric reducer
CAP
Threadolet
Socklet
Spool piece
Blind Flanges with Plug
Eccentric Reducer
Threaded Plug
Threaded CAP
Weldolet

LINE OPERATING CONDITIONS



EQUIPMENT TAGGING

MC–nn nA–nn

- SEQUENTIAL NUMBER (optional)
- SEQUENTIAL NUMBER + A or B for spare equipment (optional)
- UNIT NUMBER 2 DIGITS
- EQUIPMENT CODE
  - V: VESSEL, STORAGE TANK, OIL SEPARATOR, KNOCK OUT DRUM...
  - E: HEAT EXCHANGERS (AIR COOLED, S&T, PHE TYPE...)
  - F: FILTER
  - C: COMPRESSOR
  - F: FAN
  - P: PUMP
  - X: ELECTRIC HEATER
  - TO: THERMAL OXIDIZER
  - M.x: ELECTRIC MOTOR (x follow tag of driven machine)
  - PK: PACKAGE
  - S: SKID / SEPARATOR
  - CB: COLD BOX
  - T: COLUMN
  - TA/TK: TANK
  - B: BOILER

Rev.	Date	Description	Draft	Check'd	Approved
3	27 Mar 2026	APPROVED FOR CONSTRUCTION	TAM (JB)	TAM (MM)	I.CAMPLONE
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ManaEnergy Managem Group
Customer: MANAENERGY
OFFICES: Mahaj Riad Center, Batiment 6, 3eme etage, Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco

Contractor: ITALFLUID
OFFICES: VIA VIVALDI,4 MONTESILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742

Engineering: ITALFLUID
OFFICES: VIA VIVALDI,4 MONTESILVANO (ITALY) Tel.+39 085 4224374 fax+39 085 4220742

EPS Engineered Process Systems
Technology Vendor: TECHNICAL AMERICA INC
OFFICES: 301 N Smith Avenue, 92880 Corona, California (USA) Ph.+1 951 272 9540

TITLE: LNG TENDRARA PLANT

OBJET: P&ID LEGEND

TYPE	SCALE	DOCUMENTS ID:	VENDOR DOC. CODE	SHEET/OF
A1	N/A	PRO13–ING–PR00	21044–PR–PID–00000	1/5

LINE IDENTIFICATION

A–B–CD–E–FG

A: SIZE
1/8”, 1/4”, 3/8”, 1/2”, 3/4”, 1”, 11/4”, 11/2”, 2”, 21/2”, 3”, 4”, 5”, 6”, 8”, 10”, 12”, 14”, 16”, 18”, 20”, 24”, 30”

B: FLUID/SERVICE

AN Ammonia	AM Amine
CD Closed Drain	CW Cooling Water
EG Exhaust gas	FL Flare
GY Glycol	HC Hydrocarbon
HE Helium	IA Instrument Air
IG Instrument gas	LO Lube oil
NG Natural Gas	
NI Nitrogen	PG Process Gas
PW Process Water	RF Refrigerant
SG Seal gas	ST Steam
LG Liquified Natural Gas	BG Boil off gas
MR Mix refrigerant	HO Hot oil
PA Plant air	MH Methanol
DC Dry Condensate	WC Wet Condensate
FG Fuel gas	FW Fresh/Fire water
GO Diesel oil	RW Raw water
TO Treated oil	NL Natural gas liquid
DW Drinking/potable water	CF Cold flare
TG Treated gas	

C: nnn UNIT/SKID NUMBER

D: nn LINE NUMBERS
1 to 999 sequential line numbers from equipment

E: PIPING SPECIFICATION
1G1 Carbon Steel, Galvanized
1C1 Carbon Steel, Class 150#
3C1,3C2 Carbon Steel, Class 300#
6C1,6C3,6C5 Carbon Steel, Class 600#
9C1 Carbon Steel, Class 900#
1L1 Low Temperature Carbon Steel, Class 150#
3L1,3L3 Low Temperature Carbon Steel, Class 300#
6L1 Low Temperature Carbon Steel, Class 600#
9L1 Low Temperature Carbon Steel, Class 900#
1S1 Stainless Steel, Class 150#
3S1 Stainless Steel, Class 300#
6S1 Stainless Steel, Class 600#
9S1 Stainless Steel, Class 900#
AS1 Stainless steel, Class PN6
BS1 Stainless steel, Class PN10
CS1 Stainless steel, Class PN16

F: INSULATION TYPE (IF APPLICABLE)

P Personal Protection	C Cold
H Hot	T Heat tracing
E Electric Tracing	G Glycol Tracing

G: INSULATION THICKNESS in mm (IF APPLICABLE)

EXAMPLE:
4”–NG–200001–3C1–C50
4”: Pipe
PG: Process Gas
20001: Line 01 number unit 200
C31: Piping Specification 3C1
C50: 50mm Thick Insulation for Cold Service

VALVE TAGGING RULE

SHUT OFF MANUAL VALVE
nnn–HV–nnn
SEQUENTIAL NUMBER
UNIT/SKID NUMBER

CHECK VALVE
nnn–CK–nnn
SEQUENTIAL NUMBER
UNIT/SKID NUMBER

REGULATING MANUAL VALVE
nnn–HR–nnn
SEQUENTIAL NUMBER
UNIT/SKID NUMBER

ROOT/MANIFOLD VALVE
nnn–MV–nnn
SEQUENTIAL NUMBER
UNIT/SKID NUMBER

SPECIALTY VALVES

A B P T
Four ways hydraulic valve (spool type)– Pilot operated

A B S P T S
Four ways hydraulic valve (spool type)– Direct operated

Float expansion valve

Unidirectional Flow Control valve

PIPING INSULATION
C50

VESSEL INSULATION

PUMPS

CENTRIFUGAL PUMP

VOLUMETRIC PUMP

DIAPHRAGM PUMP

VALVE & PIPE SPECIALS SYMBOLS

Gate valve
Globe valve
Ball Valve
Butterfly valve
Needle valve
3–way valve (Gate)
4–way valve (Gate)
Check valve
Two way block and bleed manifold
Three ways block and bleed manifold
Five ways block and bleed manifold with plugs

Refr. Globe Straight Valve
Refr. Globe Angle Valve
Y Strainer
T strainer
PSV or TRV
Angle strainer
Sight glass
RO
Hole diameter
Flow Restriction Orifice
Flexible hoses

Manual regulating angle refr. valve
Manual regulating straight refr. valve
Diaphragm seal
Automatic drain

VALVE CONNECTIONS

Screwed
Flanged
Socket weld
Butt weld weld

CONTROL VALVES

DIAPHRAGM CONTROL VALVE
DIAPHRAGM CONTROL VALVE W/ POSITIONER
INLET PRESSURE REGULATOR SELF–CONTAINED
OUTLET PRESSURE REGULATOR SELF–CONTAINED
DIFFERENTIAL PRESSURE REGULATOR SELF–CONTAINED
INLET PRESSURE REGULATOR
S
2 1 3
3–WAY PILOT SOLENOID VALVE

OUTLET PRESSURE REGULATOR
HYDRAULIC/PNEUMATIC PISTON OPERATED OR SPRING RETURN
P
PNEUMATIC RELAY VALVE
S
2–WAY SOLENOD VALVE
M
MOTOR ACTUATED VALVE
ROTARY MOTOR ACTUATED
AIR LUBRICATOR
AIR REGULATOR

MISCELLANEOUS SYMBOLS

Ejector
Basket strainer
Scope of Supply
C T
CUSTOMER TG
Piping material class change
SPEC 1 SPEC 2
Scope of Electrical Installation
CUSTOMER TG
Scope of Mechanical Installation
TAM ITF
TG CUSTOMER
Skid interconnecting Tie in
An
SKID TAG
Battery limit
An
Atmospheric Seal Drain Pot
Six ports Transfer valve
Rupture disc for pressure relief
Rupture disc for vacum relief
Spectacle Blind line open
Spectacle Blind line closed
Equipment elevation above ground
XXX mm
Drip funnel
Level reference inside equipment
Conical Strainer
Bucket Strainer
Flexible hose

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A1	N/A	PRO13–ING–PROO	21044–PR–PID–00000	2/5

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

EQUIPMENT TITLE BLOCK

V-.......
VESSEL

DESIGN PRESSURE	bar g
DESIGN TEMPERATURE	°C
INTERNAL DIAMETER	mm
SM TO SM / TL TO TL	mm
CORROSION ALLOWANCE	mm
INSULATION	
LINING/CLADDING	N.A.
MATERIAL	
INTERNAL VOLUME	m3
INTERNAL ELEMENTS	

E-.........
S&T HEAT EXCHANGER

DESIGN DUTY	kW
DESIGN PRES. SHELL	bar g
DESIGN TEMP. SHELL	°C
DESIGN PRES. TUBE	bar g
DESIGN TEMP. TUBE	°C
SHELL INTERNAL DIAMETER	mm
TUBE LENGTH	mm
CORR. ALL. SHELL/TUBE	mm
INSULATION	
MATERIAL SHELL SIDE	
MATERIAL TUBE SIDE	
SURFACE AREA	m2

E-.........
PLATE HEAT EXCHANGER

DESIGN DUTY	kW
DESIGN PRES. COLD S.	bar g
DESIGN TEMP. COLD S.	°C
DESIGN PRES. HOT S.	bar g
DESIGN TEMP. HOT S.	°C
CORR. ALL. COLD/HOT	mm
INSULATION	
MATERIAL COLD SIDE	
MATERIAL HOT SIDE	
SURFACE AREA	m2

E-.........
SHELL & PLATE HEAT EXCHANGER

DESIGN DUTY	kW
DESIGN PRES. SHELL	bar g
DESIGN TEMP. SHELL	°C
DESIGN PRES. PLATE	bar g
DESIGN TEMP. PLATE	°C
SHELL INTERNAL DIAMETER	mm
SHELL LENGTH	mm
CORR. ALL. SHELL/PLATE	mm
INSULATION	
MATERIAL SHELL SIDE	
MATERIAL PLATE SIDE	
SURFACE AREA	m2

E-.........
AIRCOOLED HEAT EXCHANGER

DESIGN DUTY	kW
DESIGN PRES.	bar g
DESIGN TEMP.	°C
MATERIAL	
CORR. ALL.	mm
FAN Qty	
FAN RATED POWER each	kW
SURFACE AREA	m2

F-.......
FILTER

DESIGN PRESSURE	bar g
DESIGN TEMPERATURE	°C
INTERNAL DIAMETER	mm
SM TO SM / TL TO TL	mm
CORROSION ALLOWANCE	mm
INSULATION	
LINING/CLADDING	
MATERIAL	
INTERNAL VOLUME	m3
INTERNAL ELEMENTS	

T-.......
COLUMN

DESIGN PRESSURE	bar g
DESIGN TEMPERATURE	°C
INTERNAL DIAMETER	mm
SM TO SM / TL TO TL	mm
CORROSION ALLOWANCE	mm
INSULATION	
LINING/CLADDING	
MATERIAL (BODY)	
INTERNAL VOLUME	m3
MATERIAL INTERNALS	

X-.......
ELECTRIC HEATER

DESIGN PRESSURE	bar g
DESIGN TEMPERATURE	°C
INTERNAL DIAMETER	mm
SM TO SM / TL TO TL	mm
CORROSION ALLOWANCE	mm
INSULATION	
LINING/CLADDING	
MATERIAL (BODY)	
RATED POWER	kW
SUPPLY VOLTAGE	
SURFACE	m2

Mx-.......
ELECTRIC MOTOR

SPEED	rpm
RATED POWER	kW
VOLTAGE	Vac
FREQUENCY	Hz
STARTING	
ENCLOSURE PROTECTION	
STD	
FRAME SIZE	

C-.......
SCREW COMPRESSOR

SPEED	rpm
SWEPT VOLUME	m3/h
Vi	
MODEL	
MANUFACTURER	
CASING DESIGN PRESSURE	barg
CASING DESIGN TEMPERATURE	°C
CASING MATERIAL	

P-.......
PUMP

SPEED	rpm
RATED CAPACITY	m3/h
HEAD	bar
TYPE	
MANUFACTURER	
CASING DESIGN PRESSURE	barg
CASING DESIGN TEMPERATURE	°C
MODEL	
CASING MATERIAL	

F-.......
FAN

SPEED	rpm
RATED CAPACITY	m3/h
ABSORBED POWER	kW

X-.......
SPACE HEATER

RATED POWER	kW
SUPPLY VOLTAGE	

PK-.....
COMPRESSOR PACKAGE

SPEED	rpm
SWEPT VOLUME	m3/h
TYPE	
MODEL	
MANUFACTURER	
CASING DESIGN PRESSURE	barg
CASING DESIGN TEMPERATURE	°C

NOZZLES IDENTIFICATIONS ON VESSELS

N.. – INLETS/ OUTLETS
C.. – CONDENSATE
D – DRAIN OR DRAW OFF
L.. – LEVEL INSTRUMENT
H.. – HANDHOLES
N.. – GENERIC NOZZLE
M.. – MANHOLE
P – PRESSURE CONNECTION
S.. – SERVICE CONNECTION (PURGING,...)
T – TEMPERATURE CONNECTION
V – VAPOR OR VENT
W – RELIEF VALVE CONNECTION

TYPICAL LETTER COMBINATIONS FOR INSTRUMENT IDENTIFICATION (ISA–S5.1)

	VARIABLE	PRIMARY ELEMENT	TRANSMITTER	INDICATOR	RECORDER	CONTROLLER (BLIND)	INDICATING CONTROLLER	RECORDING CONTROLLER	CONTROL SWITCH	CONTROL ALARM	GLASS, VIEWING DEVICE	WELL (W) CONNECTION (P)	CONTROL VALVE OR REGULATOR	SELF-ACTUATED REGULATOR VALVE	SOLENOID VALVE, RELAY, CONVERTER
A	ANALYSIS	AE	AT	AI	AR	AC	AIC	ARC	AS(..)	AA(..)	–	–	AV	–	AY
B	BURNER FLAME	BE	BT	BI	BR	BC	–	–	BS(..)	BA(..)	–	–	BV	–	BY
C	CONDUCTIVITY	CE	CT	CI	CR	CC	CIC	CRC	CS(..)	CA(..)	–	–	CV	–	CY
D	DENSITY OR MASS	DE	DT	DI	DR	DC	DIC	DRC	DS(..)	DA(..)	–	–	–	–	DY
E	VOLTAGE	EE	ET	EI	ER	EC	EIC	ERC	ES(..)	EA(..)	–	–	EV	–	EY
F	FLOW (RATIO OR FRACTION)	FE	FT	FI	FR	FC	FIC	FRC	FS(..)	FA(..)	–	–	–	–	FY
G	GAUGING	GE	GT	GI	GR	GC	GIC	GRC	GS(..)	GA(..)	–	–	GV	–	–
H	HAND	–	–	–	–	HC	HIC	HRC	HS(..)	–	–	–	HV	HCV	HY
I	CURRENT	IE	IT	II	IR	IC	IIC	IRC	IS(..)	IA(..)	–	–	–	–	IY
J	POWER (SCAN)	JE	JT	JI	JR	JC	JIC	JRC	JS(..)	JA(..)	–	–	–	–	JY
K	TIME	–	KT	KI	KR	KC	KIC	KRC	KS(..)	KA(..)	–	–	–	–	KY
L	LEVEL	LE	LT	LI	LR	LC	LIC	LRC	LS(..)	LA(..)	LG	–	LV	LCV	LY
M	MOISTURE,HUMIDITY	ME	MT	MI	MR	MC	MIC	MRC	MS(..)	MA(..)	MG	–	MV	–	MY
P	PRESSURE / VACUUM	–	PT	PI	PR	PC	PIC	PRC	PS(..)	PA(..)	PG	–		PCV (**)	PY
PD	PRESSURE DIFFERENTIAL	PDE	PDT	PDI	PDR	PDC	PDIC	PDRC	PDS(..)	PDA(..)	PDG	–	PDV	PDCV	PDY
Q	QUANTITY OR EVENT (INTEGRATE/TOTALIZE)	–	QT	QI	QR	QC	QIC	QRC	QS(..)	QSA(..)	–	–	QV	–	QY
R	RADIOACTIVITY	RE	RT	RI	RR	RC	RIC	RRC	RS(..)	RSA(..)	–	–	–	–	RY
S	SPEED OR FREQUENCY	–	ST	SI	SR	SC	SIC	SRC	SS(..)	SSA(..)	–	–	–	–	SY
T	TEMPERATURE	TE	TT	TI	TR	TC	TIC	TRC	TS(..)	TSA(..)	TG	TW	TV	TCV	TY
TD	TEMPERATURE DIFFERENTIAL	TDE	TDT	TDI	TDR	TDC	TDIC	TDRC	TDS(..)	TDA(..)	–	–	TDV	TDCV	TDY
V	VIBRATION	VE	VT	VI	VR	VC	–	–	VS(..)	VA(..)	–	–	–	–	VY
W	WEIGHT OR FORCE	WE	WT	WI	WR	WC	–	–	WS(..)	WA(..)	–	–	WV	–	WY
X	UNCLASSIFIED DIAGNOSTIC/CONTROL	XE	XT	XI	XR	XC	–	–	XS(..)	XA(..)	–	–	–	–	XY
Z	POSITION	ZE	ZT	ZI	ZR	ZC	ZIC	ZRC	ZS(..)	ZA(..)	–	–	–	–	ZY

(*) RO: RESTRICTION ORIFICE
(**) PSV: PRESSURE RELIEF OR SAFETY VALVE
(***) XL: LIGHTS or STATUS SIGNAL ON HMI
(****) PL: PANEL LIGHT

(C)–CLOSE (O)– OPEN
(H)– HIGH ALARM (L)– LOW ALARM
(HH)– HIGH SHUTDOWN (LL)–LOW SHUTDOWN

(XX)– DIAGNOSTIC SHUTDOWN
(USED TO INDICATE THE DIAGNOSTIC CHECK REQ'D ON THE ANALOG INPUT)

INSTRUMENT SYMBOLS

LOCAL INSTRUMENT

INACCESSIBLE INSTRUMENT LOCAL MOUNTED

INSTRUMENT / PUSH BUTTON LOCAL PANEL MOUNTED

CONTROL ROOM (HMI)

CONTROL ROOM INTERLOCK (MAIN PLANT PLC)

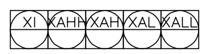
LOCAL PLC INTERLOCK (PACKAGE PLC)

INTERLOCK LOGIC Fire & Gas

Motor Control Cabinet

COMPLEX FUNCTION

SYNTHETIC – EXTENDED REPRESENTATIONS



MANUAL RESET

PILOT LIGHT ON LOCAL PANEL

X DESIGNATION:
R – RED
G – GREEN
B – BLUE
W – WHITE
A – AMBER

ROTAMETER

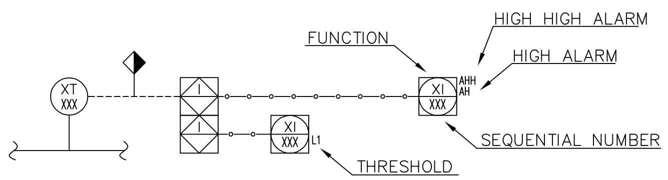
MAGNETIC FLOWMETER

CORIOLIS FLOWMETER

THERMAL MASS FLOWMETER

PILOT LIGHT IN CONTROL ROOM (HMI)

INSTRUMENTS REPRESENTATION ON P&ID



AHH – HIGH HIGH ALARM
AH – HIGH ALARM
H1 – THRESHOLD
H2 – THRESHOLD
... – ...
... – ...
... – ...
Hn – THRESHOLD
Ln – THRESHOLD
... – ...
... – ...
... – ...
L2 – THRESHOLD
L1 – THRESHOLD
AL – LOW ALARM
ALL – LOW LOW ALARM

LOOP NUMBER

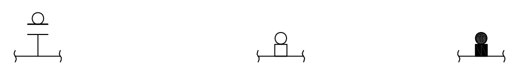
.nn .nn
SEQUENTIAL NUMBER
UNIT NUMBER 1st + 2nd digits

THERMOWELL

Flanged Thermowell with threaded well

Threaded Thermowell

Weld in Thermowell



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Technology Vendor: TECHNICAL AMERICA INC
OFFICES: 301 N Smith Avenue, 92880 Corona, California (USA) Ph.+1 951 272 9540

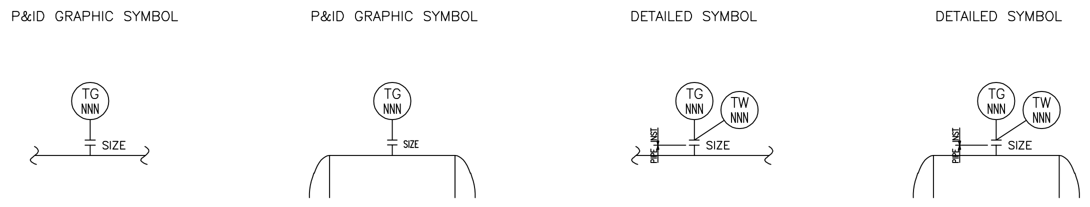
TITLE: LNG TENDRARA PLANT

OBJET: P&ID LEGEND

TYPE	SCALE	DOCUMENTS ID:	VENDOR DOC. CODE	SHEET/OF
A1	N/A	PRO13–ING–PROO	21044–PR–PID–00000	3/5

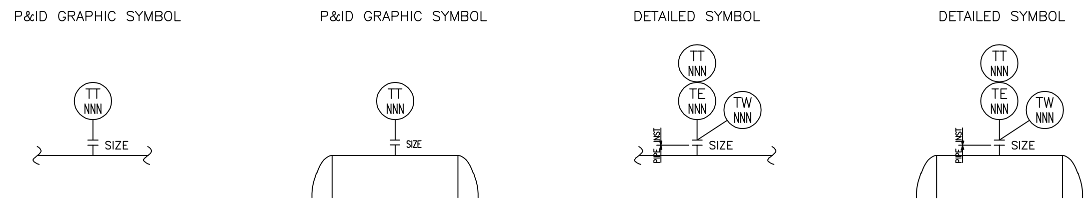
TYPICAL INSTRUMENT INSTALLATIONS

TEMPERATURE GAUGE TG (LOCALLY MOUNTED)



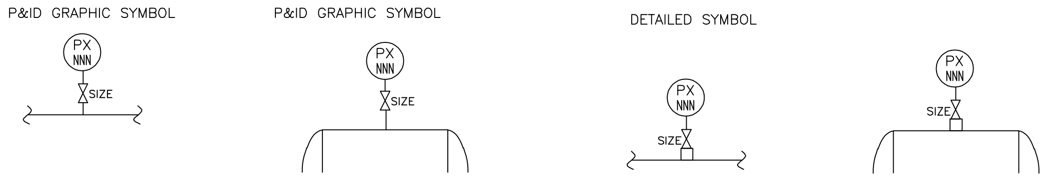
THERMOWELL CAN BE THREADED, WELDED OR FLANGED

TRANSMITTER TT (LOCALLY MOUNTED)



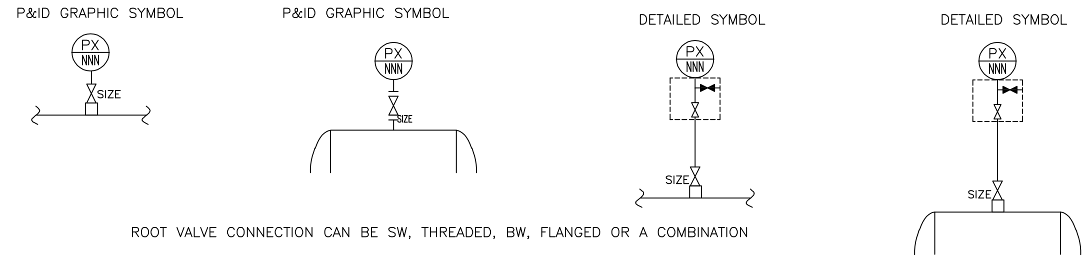
THERMOWELL CAN BE THREADED, WELDED OR FLANGED

TRANSMITTER PT OR GAUGE PG (LOCALLY MOUNTED)



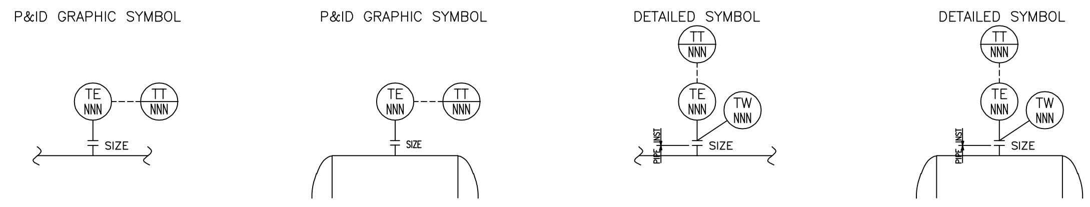
ROOT VALVE CONNECTION CAN BE SW, THREADED, BW, FLANGED OR A COMBINATION

TRANSMITTER PT OR GAUGE PG (REMOTE MOUNTED)



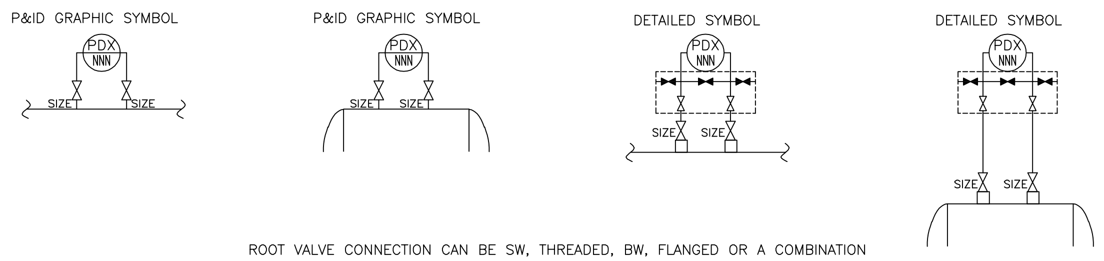
ROOT VALVE CONNECTION CAN BE SW, THREADED, BW, FLANGED OR A COMBINATION

TRANSMITTER TT (REMOTE MOUNTED)



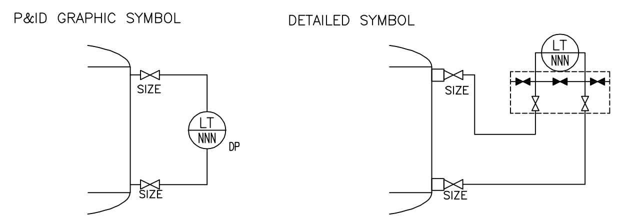
THERMOWELL CAN BE THREADED, WELDED OR FLANGED

DIFFERENTIAL PRESSURE GAUGE PDG AND TRANSMITTER PDT



ROOT VALVE CONNECTION CAN BE SW, THREADED, BW, FLANGED OR A COMBINATION

LEVEL TRANSMITTER DP TYPE ON VESSEL

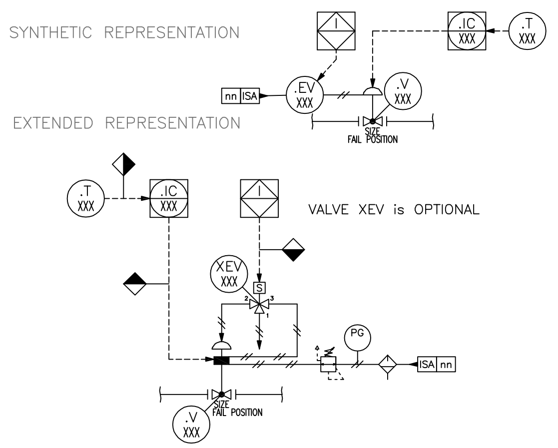


ROOT VALVE CONNECTION CAN BE SW, THREADED, BW, FLANGED OR A COMBINATION

TYPICAL N° 1 (PNEUMATIC CONTROL VALVE)

SYNTHETIC REPRESENTATION

EXTENDED REPRESENTATION

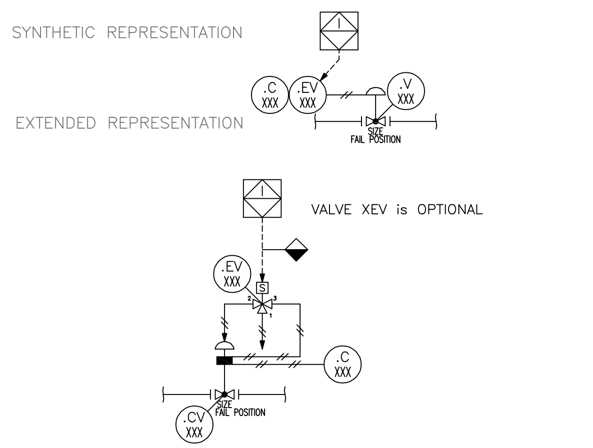


VALVE XEV is OPTIONAL

TYPICAL N° 2 (PNEUMATIC CONTROL WITH PNEUMATIC CONTROLLER)

SYNTHETIC REPRESENTATION

EXTENDED REPRESENTATION

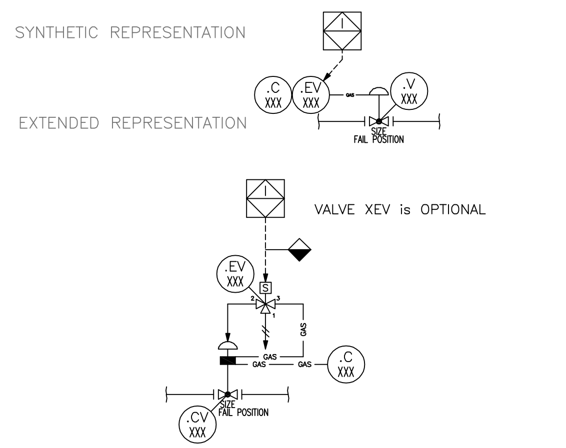


VALVE XEV is OPTIONAL

TYPICAL N° 3 (GAS ACTUATED CONTROL WITH GAS CONTROLLER)

SYNTHETIC REPRESENTATION

EXTENDED REPRESENTATION

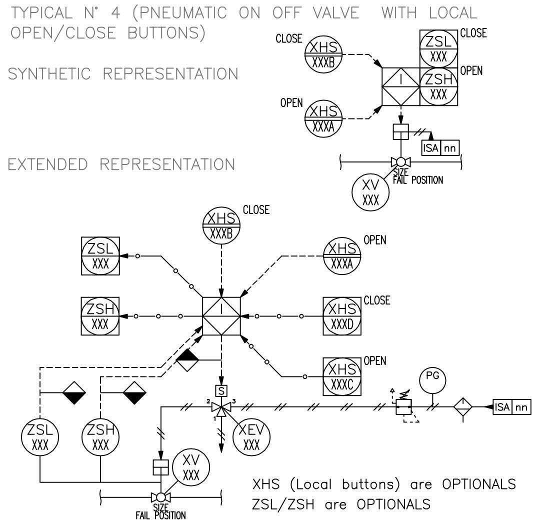


VALVE XEV is OPTIONAL

TYPICAL N° 4 (PNEUMATIC ON OFF VALVE WITH LOCAL OPEN/CLOSE BUTTONS)

SYNTHETIC REPRESENTATION

EXTENDED REPRESENTATION



XHS (Local buttons) are OPTIONALS
ZSL/ZSH are OPTIONALS

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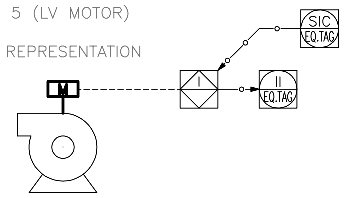
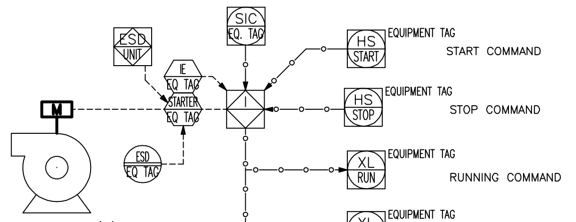
TYPICAL MOTORS AND OTHER ELECTRIC USER INSTALLATION

TYPICAL N° 5 (LV MOTOR)

SYNTHETIC REPRESENTATION

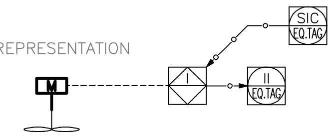


EXTENDED REPRESENTATION

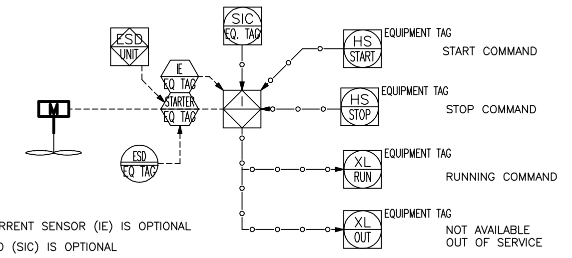


CURRENT SENSOR (IE) IS OPTIONAL
VFD (SIC) IS OPTIONAL

SYNTHETIC REPRESENTATION



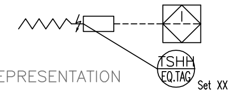
EXTENDED REPRESENTATION



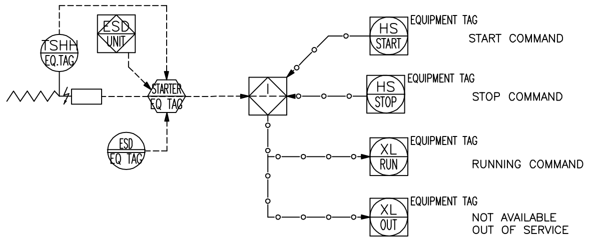
CURRENT SENSOR (IE) IS OPTIONAL
VFD (SIC) IS OPTIONAL

TYPICAL N° 6 (ELECTRIC HEATER)

SYNTHETIC REPRESENTATION



EXTENDED REPRESENTATION



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